

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Any two (x1) from - Many {alveoli/capillaries} / folded alveoli + {provide a large surface area/ increasing <u>rate</u> of diffusion} (1) {Squamous epithelium/alveolar walls/ capillary walls} are {thin/ one cell thick} + {short diffusion pathway/ increasing <u>rate</u> of diffusion} (1) Reject thin cell walls {Many capillaries/ good blood supply/ capillary network} + {maintain (steep) concentration gradients/ increasing <u>rate</u> of diffusion} (1)	2			2		2
		(ii)		(Breakdown of the alveolar walls) reduces the surface area (of the alveoli) (1) - Less oxygen <u>diffuses</u> into the blood/ decreases the rate of <u>diffusion</u> (of oxygen) into the blood (1)		1	1	2		2
	(b)	(i)		Any four (x1) from (External) Intercostal muscles contract and the <u>rib</u> cage moves {upwards / outwards / expands} (1) - The diaphragm contracts and {flattens/ goes down} (1) - Pulls on pleural membranes (1) - Increased {thoracic/ thorax/ lung/ alveolar} volume (1) - Which decreases {lung/alveolar} pressure (1)	4			4		